GigaPet Chrome DevTools Report

leviathan516.github.io/HW10-Chrome-Dev-Tools/.

1. Message Logging

Log Info, Warning, Error

Helper functions in script.js are called on every state update.

```
console.info('[GigaPet]      Action:',      action,      '|      State:',      pet_info);
console.warn('[GigaPet]      Energy      critically      low!',      pet_info.energy);
console.error('[GigaPet] Stat hit boundary.', value, limit);
```

Log Table

egg.js logs the pets roster as a sortable table on page load.

```
console.table(pets);
```

Log Group

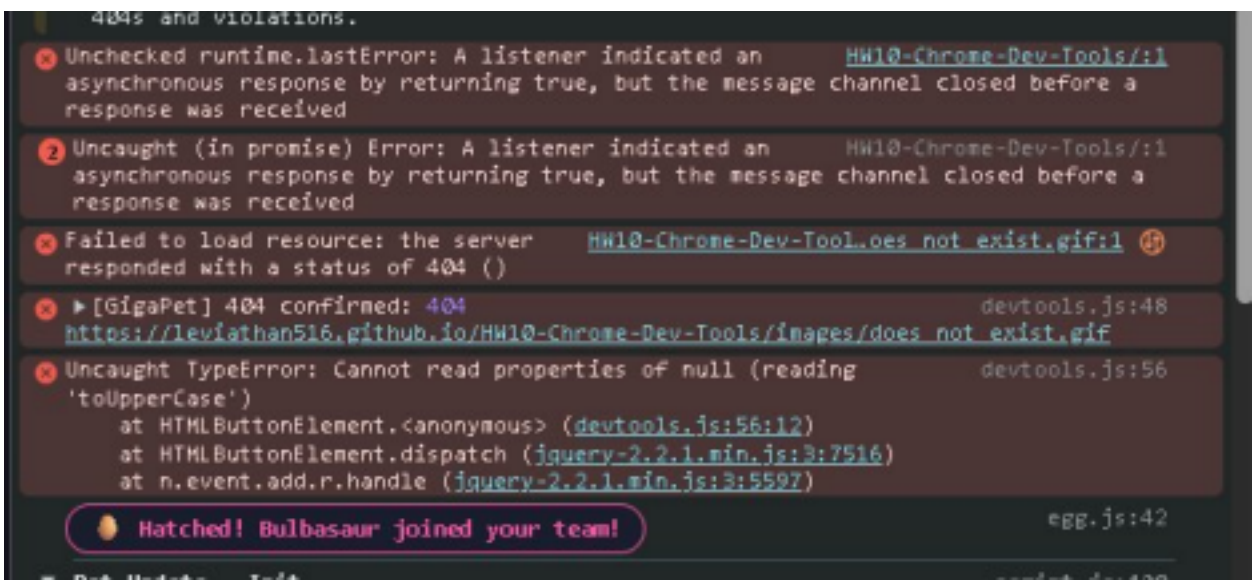
script.js wraps each update cycle in a collapsible group.

```
console.group('Pet Update -' + action); // ...logs... console.groupEnd();
```

Log Custom (%c)

Styled banners on page load and pet hatch -see Screenshot 1.

```
console.log('%c■ Hatched! ' + chosenPet.name, 'background:#1a1f2e; color:#ff4da6;
padding:4px 12px;' + 'border-radius:20px; border:1px solid #ff4da6;');
```



2. Browser-Logged Messages

A floating demo panel (devtools.js) has three buttons that trigger browser errors.

404 Network Error

```
fetch('images/does_not_exist.gif');
```

TypeError

```
var broken = null; broken.toUpperCase(); // throws TypeError
```

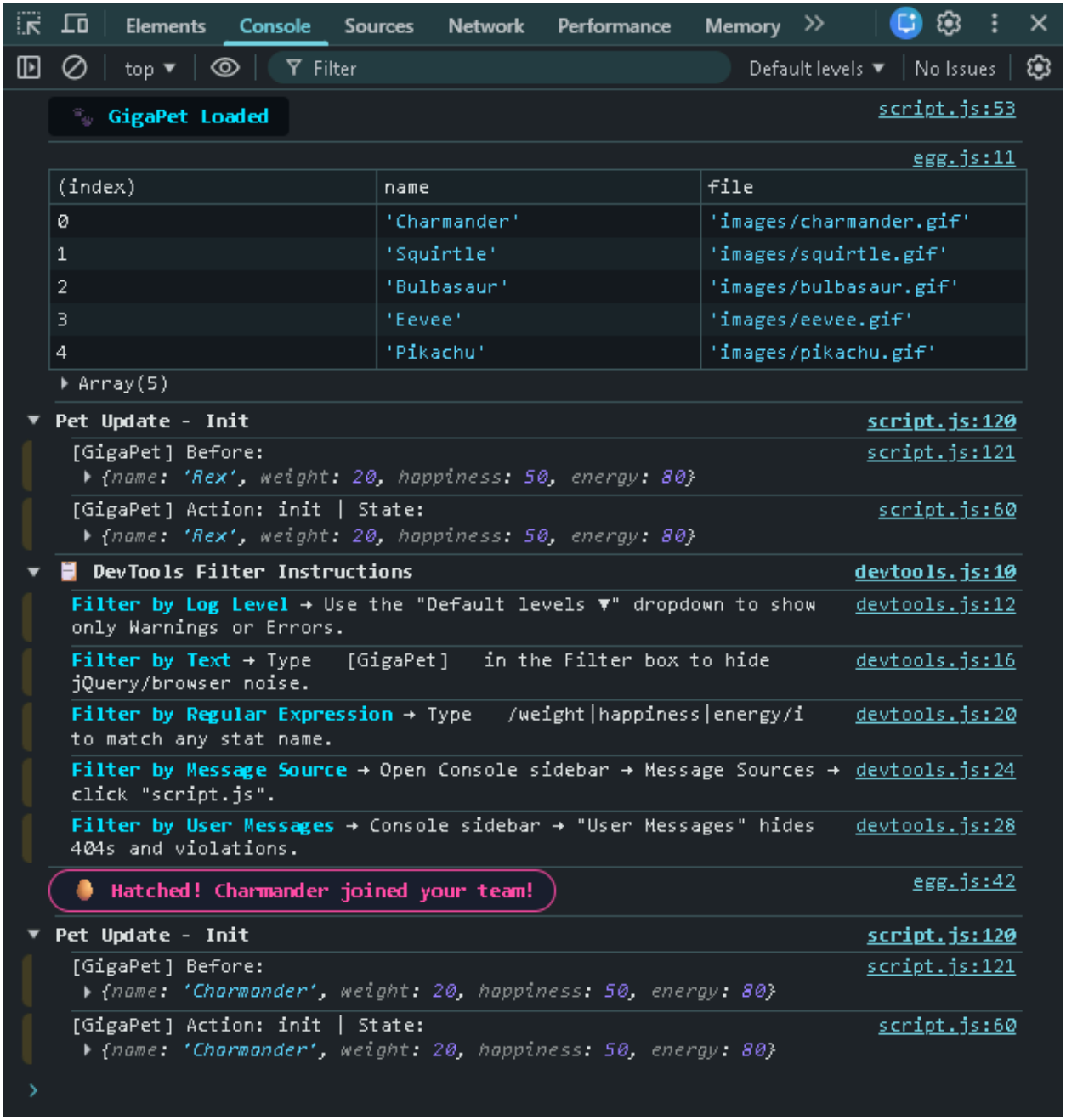
Violation

```
var start = Date.now(); while (Date.now() - start < 350) { /* busy-wait */ }
```

3. Filter Messages

All five filter mechanisms are documented in console output on page load:

```
By log level → Default levels ▼ → Warnings / Errors By text → type [GigaPet] By regex
→ type /weight|happiness|energy/i By source → Sidebar → Message Sources → script.js
By user msgs → Sidebar → User Messages
```



4. Reproduce a Bug

Bug: Clicking Treat increased happiness by 20 instead of 10. The Console showed two consecutive 'Pet Update' groups per click instead of one.

To reproduce:

• Open GigaPet in Chrome and open DevTools (F12).

• Switch to the Console tab and clear any existing output.

• Pick any egg to advance to the game screen.

• Click the Treat button once.

• Observe two 'Pet Update, Treat' groups in the Console.

• Observe happiness changed from 50 → 70 instead of 50 → 60.

Root Cause:

jQuery's .click() appends handlers; it never replaces them. With identical bindings in two ready handlers, every button accumulated two listeners.

```
$(function() {
  buildEggScreen();

  // $('.treat-button').click(clickedTreatButton);     // removed (duplicate)
  // $('.play-button').click(clickedPlayButton);       // removed (duplicate)
  // $('.exercise-button').click(clickedExerciseButton); // removed (duplicate)
  // $('.rest-button').click(clickedRestButton);       // removed (duplicate)
});
```

```
function clickedTreatButton() {
  pet_info.happiness += 10;   // Set line-of-code breakpoint here
  pet_info.weight    += 2;
  showSpeechBubble("Yum! That was delicious! &#x1F36A;");
  checkAndUpdatePetInfoInHtml('Treat');
}

function clickedPlayButton() {
  pet_info.happiness += 8;
  pet_info.weight    -= 1;
  pet_info.energy    -= 10;
  showSpeechBubble("Woohoo! Let's play all day! &#x1F3BE;");
  checkAndUpdatePetInfoInHtml('Play');
}

function clickedExerciseButton() {
  pet_info.happiness -= 5;
  pet_info.weight    -= 3;
  pet_info.energy    -= 15;
  showSpeechBubble("I'm tired... but I feel stronger! &#x1F4AA;");
  checkAndUpdatePetInfoInHtml('Exercise');
}

function clickedRestButton() {
  pet_info.energy    += 30;
  pet_info.happiness -= 3;
  showSpeechBubble("Zzz... that nap was great! &#x1F634;");
  checkAndUpdatePetInfoInHtml('Rest');
}

function checkAndUpdatePetInfoInHtml(action) {
  // Uncomment the next line for a programmatic breakpoint
  // debugger;

  console.group('Pet Update - ' + (action || 'Init'));
  console.info('[GigaPet] Before:', JSON.parse(JSON.stringify(pet_info)));

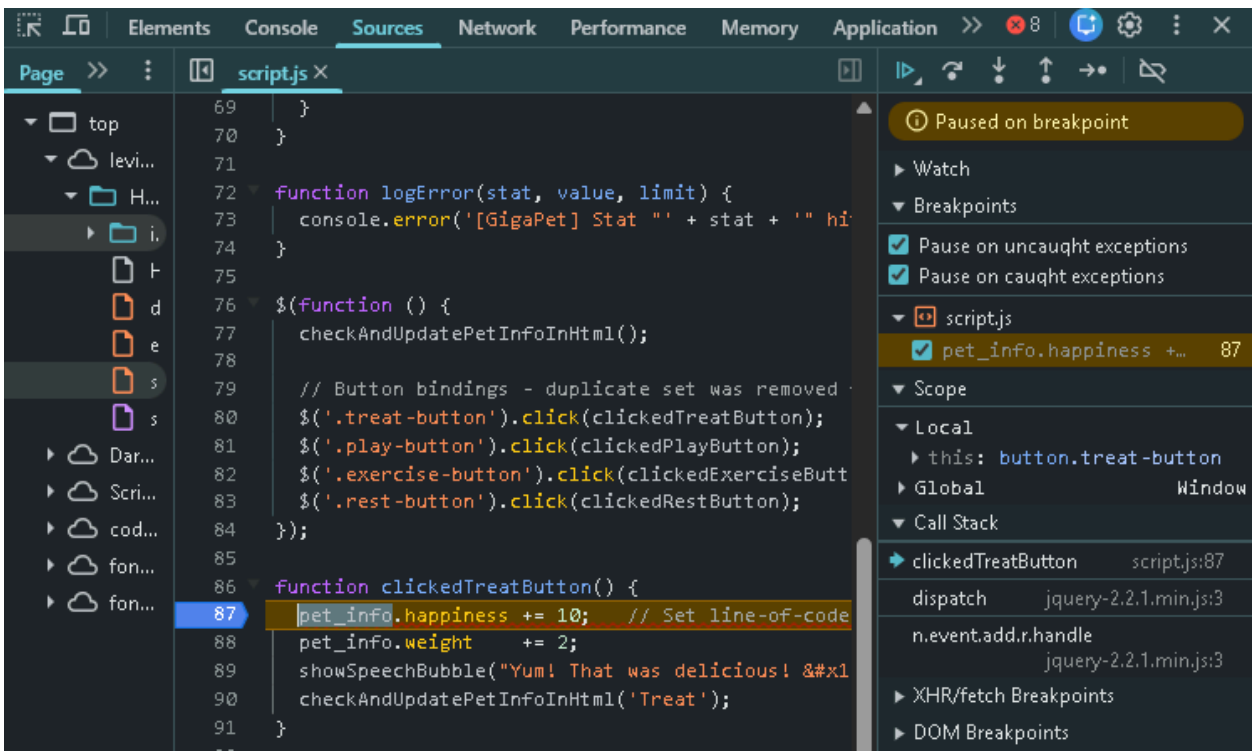
  checkWeightAndHappinessBeforeUpdating();
  updatePetInfoInHtml();
  updatePetImageStyle();
```

5. Sources UI & Breakpoints

The Sources panel has three panes: File Navigator (left), Code Editor (center), Debugger Pane (right) with Resume, Step Over, Step Into, Step Out controls.

Set a Breakpoint

Open script.js (Ctrl+P), click the gutter line number for **pet_info.happiness += 10;** inside clickedTreatButton(). Click Treat -execution pauses before that line runs.



6. Check Variable Values

Scope Pane

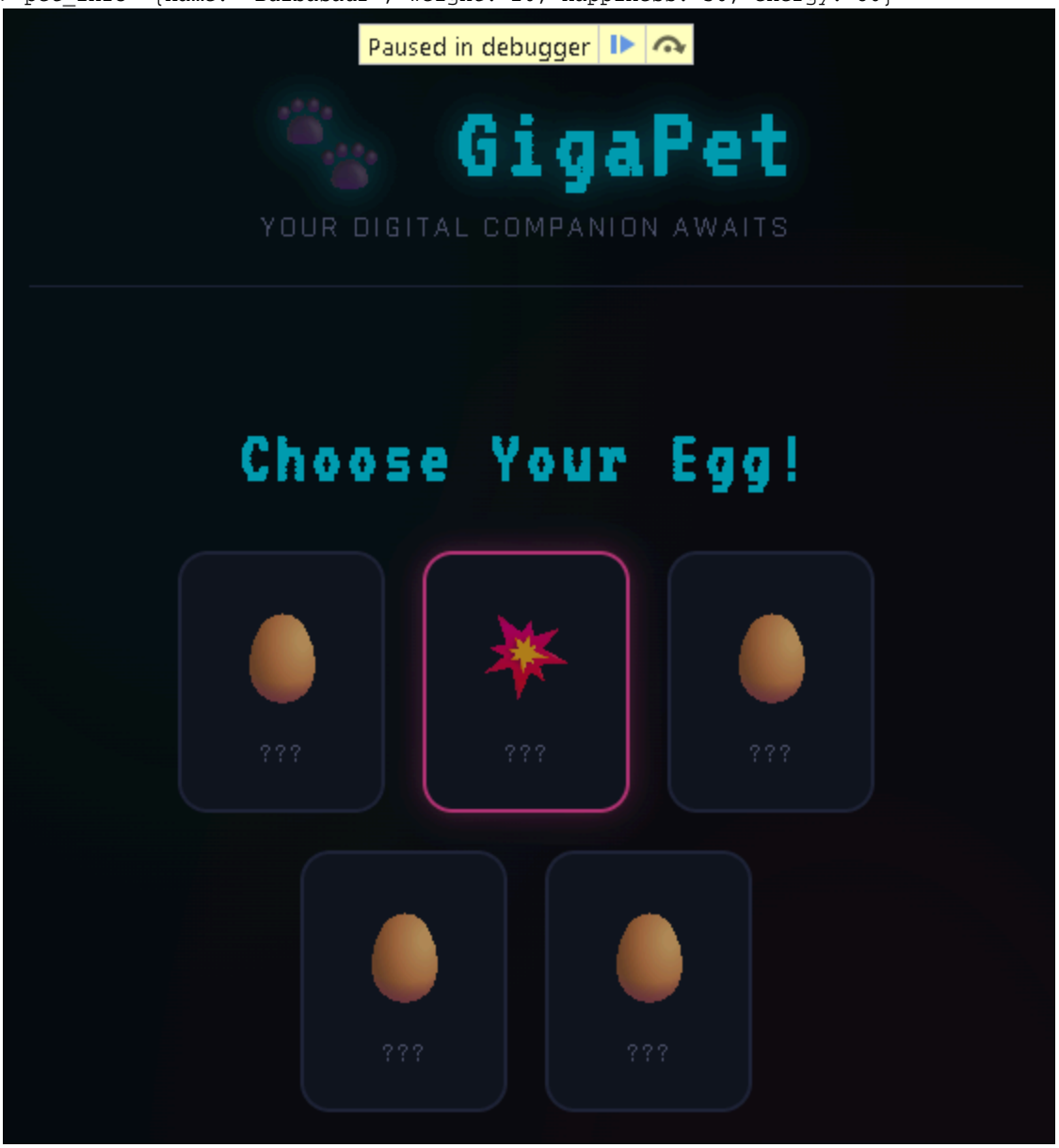
When paused, expand 'Global' to see pet_info. Inside hatchEgg(), the local **chosenPet** appears under 'Local'.

Watch Expressions

```
pet_info.happiness pet_info.happiness + 10 pet_info.energy < 20
```

Console (while paused)

```
> pet_info -{name: 'Bulbasaur', weight: 20, happiness: 50, energy: 80}
```



```
$egg.find('.egg-icon').text('');

setTimeout(function() {
  var chosenPet = $egg.data('pet');   // local var visible in Scope pane
  pet_info.name = chosenPet.name;
  $('.pet-image').attr('src', chosenPet.file);

  // Custom styled batch banner — console.log with %c
```

7. Apply a Fix

Removed the four duplicate .click() bindings from egg.js. Bindings now live in script.js only. After fix: one 'Pet Update -Treat' group per click; happiness changes by exactly +10.

```
// egg.js -fixed $(function() { buildEggScreen(); // Duplicate bindings removed
-handlers live in script.js });
```

```
$(function() {
  buildEggScreen();

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